4.2 Building Trigonometry from an Integral

WORKSHEET

The question

STEP III 2017, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Which trigonometric identities do parts (i) and (ii) correspond to? (Think of $\arctan x + \arctan \frac{1}{x}$, and the formula for $\tan(A+B)$.)

STAGE 1 The reflection $u \mapsto 1/u$

Covers: Part (i)

Pause and try Substitute $u = 1/w$ in $\mathrm{T}(x)$. What happens to the integrand, and to the limits?

STAGE 2 The addition substitution

Covers: Part (ii)

Ask yourself Differentiate $v = \frac{u+a}{1-au}$ by the quotient rule. Then write $1 + v^2$ as a single fraction and compare.

Pause and try So $\frac{du}{1+u^2} = \frac{dv}{1+v^2}$: the substitution does not change the integrand. What does it do to the limits of $\mathrm{T}(x)$?

STAGE 3 Two more identities

Covers: Part (ii)

Ask yourself Use part (i) twice to replace $\mathrm{T}(x)$ and $\mathrm{T}(a)$. Then rename: put $y = 1/x$ and $b = 1/a$. What does the condition $x < 1/a$ become?

STAGE 4 Special values

Covers: Part (iii)

Pause and try For $\mathrm{T}(\sqrt{3})$: choose y and b in the last identity so that $\frac{y+b}{by-1}$ equals y itself.

Ask yourself For $\mathrm{T}(\sqrt{2}-1)$: use the addition identity with $x = a = \sqrt{2}-1$. What is $\frac{2x}{1-x^2}$? And what does part (i) give for $\mathrm{T}(1)$?

What's really going on

Building the elementary functions

In a rigorous analysis course, the functions exp, sin, cos and ln cannot simply be assumed; they have to be *constructed*, and their properties proved. There are three standard approaches. You can define them by power series ($\sin x = x - \frac{x^3}{3!} + \cdots$). You can define them as solutions of differential equations, as in the 1986 specimen question in the exercises ($\mathrm{f}' = \mathrm{f}$, $\mathrm{f}(0) = 1$ defines exp). Or you can define inverse functions by integrals, as here: $\ln x = \int_1^x \frac{\mathrm{d}u}{u}$ and $\arctan x = \int_0^x \frac{\mathrm{d}u}{1+u^2}$, and then define $\pi = 2\int_0^\infty \frac{\mathrm{d}u}{1+u^2}$. Each approach has its strengths, and proving that they all give the same functions is a satisfying exercise. Compare the functional-equation approach of Section 3.8.

Beyond the circle: elliptic integrals

The integral approach generalises in a way the triangle approach cannot. In 1718 Giulio Fagnano studied the arc length of the *lemniscate*, a figure-of-eight curve, which leads to $\int_0^x \frac{du}{\sqrt{1-u^4}}$, and found a substitution that doubles the arc, just as our substitution doubled T. Euler, reading Fagnano's work in 1751, found a full addition formula, and this led eventually, through Abel and Jacobi, to the theory of *elliptic functions*: inverse functions of integrals like this one, which are to the lemniscate what sin and cos are to the circle. We met the same integral in Gauss's work on the AGM (Section 2.6). Elliptic functions are now central to number theory; the elliptic curves of Section 1.9 are their geometric shadow.

Computing π

The addition identity is also the basis of the most famous formulas for computing π. Repeatedly applying it shows, for instance, that $\mathrm{T}(1) = 4\mathrm{T}(\frac{1}{5}) - \mathrm{T}(\frac{1}{239})$, which is Machin's formula of 1706; combined with the series $\mathrm{T}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, it gave 100 digits of π and stayed the method of choice for two and a half centuries. That is the subject of Section 5.1.

Try this next

1. **The exponential and the cosine.** Try STEP II Specimen paper, Question 6 (step.maths.org/questions/spec-s2-q6). It derives the addition formulae for the exponential and the cosine from their differential equations alone. (*Hint:* for the cosine, find two combinations that ought to be zero, and show that the sum of their squares is constant.)

2. **The inverse sine.** Try STEP III 2007, Question 7 (step.maths.org/questions/07-s3-q7). It gives the inverse sine the same treatment as this section's integral, and links the two by a substitution.

3. **An early Machin-type formula.** Use the addition identity to show that $\mathrm{T}(\frac{1}{2}) + \mathrm{T}(\frac{1}{3}) = \mathrm{T}(1) = \frac{1}{2}\mathrm{T}_\infty$.

4. **T_∞ is finite.** Show that $\mathrm{T}(x) \le 1 + \int_1^x \frac{du}{u^2} < 2$ for all $x \ge 1$, explain why $\mathrm{T}(x) < 2$ for $0 < x < 1$ as well, and deduce from the monotone convergence theorem (Section 2.2) that T_∞ exists.

Further reading

- **Tom Körner, *Calculus for the Ambitious*** (Cambridge University Press). Builds the elementary functions rigorously, discussing the different approaches.
- **Walter Rudin, *Principles of Mathematical Analysis*** (McGraw-Hill). The famous "baby Rudin"; the prologue to Chapter 8 constructs exp, sin and cos from power series in

a few elegant pages. Advanced.

- **Victor Prasolov & Yuri Solovyev, *Elliptic Functions and Elliptic Integrals*** (American Mathematical Society). For where the lemniscate leads. Advanced.

The full guided solution is in Section 4.2:
https://beyondstep.keithmcnulty.org/ch04/trig-from-integral.html